

Aritra Dutta

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Education

Jadavpur University, Kolkata

2024 – Present

B.E. in Computer Science and Engineering

- **Grade:** 9.51 CGPA

H.M. Education Centre, Hindmotor

2009 – 2024

Higher Secondary (XII)

- **Grade:** 97.8% (AISSCE) & 98.2% (CBSE 10th)

Programming Achievements

- **LeetCode:** Guardian | Max Rating: 2155 | Top 1.16% globally | [aricode25](#)
- **Codeforces:** Specialist | Max Rating: 1620 | [aricode2005](#)
- **CodeChef:** 3-Star | Max Rating: 1656 | [aricode25](#)

Technical Proficiencies

Languages: C, C++, Java, Python

Database Tools: My SQL, Mongo DB

Core CS: Computer Organisation, DSA, OOPs, DBMS, Computer Networks, Operating Systems

Web Frameworks: Node.js, React.js, Express.js, Rest APIs

Agentic AI Tools: LangChain, LangSmith, LangGraph

Tools/Others: Docker, Apache Kafka, Redis, Git, GitHub

Projects

NexusPay Engine (Enterprise FinTech API) | *Node.js, PostgreSQL, Apache Kafka, Redis, Hugging Face* [GitHub](#) | July 2026

- Built an event-driven digital wallet backend using **Apache Kafka** to decouple ledger updates from notifications, reducing API response latency by an estimated **40%**.
- Designed a Two-Step Intent/Execute flow with **PostgreSQL** row-level locks and UUID-based ordering, ensuring **ACID compliance** while eliminating deadlocks and double-spend race conditions.
- Secured transaction endpoints with **Redis**-based rate limiting, idempotency keys, and **JWT**-based stateless authentication, blocking brute-force MPIN attacks and duplicate retries.
- Built an **Agentic AI Fraud Detection** system that monitors real-time transaction velocity and autonomously freezes suspicious accounts, plus a **RAG** Financial Assistant that lets users query past transactions in natural language.

ExpenseIQ (Agentic AI FinTech Server) | *Python 3.13, FastMCP, aiosqlite, SQLite* [GitHub](#) | Sept 2026

- Developed a production-grade backend using the **Model Context Protocol (MCP)**, safely exposing 15 complex financial tools (CRUD, budget tracking, analytics) directly to LLMs like Claude.
- Engineered a fully asynchronous architecture with **aiosqlite** for non-blocking I/O, and successfully deployed it to a remote cloud environment via **Streamable HTTP** transport.
- Designed fault-tolerant natural language workflows where the AI can autonomously validate inputs, self-correct categorization errors, and instantly access live financial dashboards via MCP Resources.

QuantStream Engine (Algorithmic Trading Backtester) | *C++17, OOPs* [GitHub](#) | March 2026

- Built a high-performance **C++17** algorithmic trading engine leveraging **OOP** principles (polymorphism) to seamlessly swap 5+ statistical strategies (RSI, Bollinger Bands, Z-Score) without modifying core execution logic.
- Implemented shared risk-management modules to drastically reduce code duplication, and engineered advanced post-trade diagnostics like the **Hurst Exponent** for dynamic market-regime detection.

Ranks & Achievements

- **WBJEE 2024:** Achieved a General Merit Rank (GMR) of **73** among 98K students
- **JEE Advanced 2024:** Achieved an All India Rank (AIR) of **4229** among 2.5 lakh students
- **JEE Mains 2024:** Achieved an All India Rank (AIR) of **5165** among 1.1 million students
- **Second Runner Up at OpenAimer Srijan'26**, Code Club JU, Kolkata
- **Meta Hacker Cup 2025:** Ranked 2711th out of 13,779 globally in Round 1 (Human Track), Qualified to Round 2
- **Apache Kafka**, Udemy